

Noah Bean

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EDUCATION

Oregon State University – Corvallis, OR
B.S. Electrical and Computer Engineering

Dec 2025
GPA: 3.54/4.0

TECHNICAL SKILLS

Systems Software: C, C++, Python, PyTorch, x86 Assembly, Bash, Make, Linux, Intel VT-x, Docker, Git, GDB
Core Concepts: Operating Systems, Computer Architecture, Virtualization, Parallel Programming, Computer Networking, Hardware/Software Co-Design

EXPERIENCE

Systems Engineering Co-op – Collins Aerospace, Wilsonville, OR

Jun 2025 - Dec 2025

- Developed **validation automation tools** in Python for data collection and visualization, improving test efficiency for RF and infrared avionics systems.
- Developed real-time signal processing algorithms for embedded hardware optimization based on field and flight testing data.
- Evaluated system modifications through rigorous validation protocols, ensuring compliance with standards.

Electrical Engineering Intern – Intel Corporation, Hillsboro, OR

Apr 2024 – Sep 2024

- Debugged server and GPU hardware failures down to the signal level using oscilloscopes and logic analyzers, reducing system downtime.
- Engineered a **PyTorch-based neural network** for anomaly detection, automating the identification of manufacturing defects.
- Developed Python automation scripts for scan-chain design and simulation.

PROJECTS

Hypervisor – C, x86 Assembly, WDK

2025

- Developed a kernel-mode driver utilizing **Intel VT-x** to virtualize the running operating system and intercept hardware events.
- Implemented **VMCS** (Virtual Machine Control Structure) lifecycles and **Extended Page Tables (EPT)** to manage guest physical memory translation.
- Engineered VMX root/non-root mode switching logic to handle VM-exits and monitor guest execution state.

Multithreaded Kernel – C, x86 Assembly

2025

- Built a 32-bit protected mode kernel featuring a custom round-robin scheduler for preemptive multitasking and privilege separation (**Ring 0 vs Ring 3**).
- Implemented virtual memory paging, a programmable interrupt controller (PIC) driver, and a FAT16 filesystem driver.
- Integrated an ELF binary loader to execute user-space programs and manage system calls for process isolation.

C Compiler – C, Make

2025

- Engineered a self-hosting C compiler from scratch, implementing a custom lexer, AST parser, and x86 assembly code generator.
- Validated functionality against GCC standards for pointer arithmetic, struct memory alignment, and stack frame management.
- Optimized semantic validation for control flow structures (loops/conditionals) and primitive data types.